DIAMOND VS COPPER

Parameter	Diamond	Copper
Thermal conductivity	Around 5× higher than copper	Lower than diamond
Heat spreading	Moves heat quickly away from small hotspots and spreads it over a larger area	Effective for general heat spreading but less effective at very high heat densities
Best suited for	High-power packages with concentrated heat	Conventional thermal management
Bonding	May require metallisation, brazing, or specialised adhesive layers as diamond does not naturally bond well with some package materials	Easier to integrate with commonly used package materials
Bonding impact	A thermally resistive bonding layer can reduce the benefit of diamond	Existing bonding materials are generally easier to use
Cost	Higher than copper, aluminium, and graphite	Lower cost
Integration	More complex and may require additional processing	Relatively simple to integrate
Key consideration	Higher cost and integration complexity need to be justified by the thermal benefits	Lower cost and easier integration make it suitable for many conventional applications

Table A

as electronic heat spreaders.

But what makes diamond such an effective thermal material in the first place? The answer lies in its atomic structure. Carbon atoms form an orderly structure connected by strong bonds, letting heat-related vibrations travel quickly through the material. Its low atomic mass also helps these vibrations move even faster, and in high-quality diamond, fewer crystal defects mean fewer interruptions along the way.

That does not mean diamond is about to replace copper everywhere. Cost and the engineering effort needed to integrate it into a package still count against it. For now, its strongest case is in designs where heat has become the binding constraint, and where the thermal payoff is worth the trade-offs.

Fitting diamonds inside the package

Diamond is finding a place inside electronic packages mainly as a thermal management material. Adding it doesn't always mean bolting on an entirely new layer. Depending on the design, it can replace part or all of an existing heat spreader, sit closer to the chip, or become part of the package base itself.

Take a typical package stack: printed circuit board (PCB), substrate, silicon chip, thermal interface material (TIM), copper heat spreader, a second TIM layer, and a cold plate. Diamond can be introduced at several points in this structure. One of the more direct approaches places a diamond plate on top of the chip, separated by thin bonding layers. The diamond sits close to the heat source, where it can pull heat from small hot spots and spread it over a wider area before it reaches the main cooling system.

Diamond can also replace some or all of the copper heat spreader, be bonded underneath a copper lid, or embedded inside the lid to form a diamond-copper structure. Copper-diamond composites are another option, offering thermal conductivity of roughly 800W/m·K, giving designers a choice between single-crystal diamond, polycrystalline diamond, or a composite, depending on the package. It can sit directly below a semiconductor too. In a GaN device, for instance, a diamond submount can go beneath the device and above a copper base plate. Another configuration builds diamond into the package base itself, with a power semiconductor on a ceramic substrate, followed by a diamond-copper base plate and a cold plate. In these layouts, diamond can substitute for copper, aluminium, or silicon carbide, or work alongside them as part of the thermal path.

The benefits are most pronounced where a device produces a lot of heat in a small area. GaN RF flip-chip and diode packaging are strong examples—GaN devices used in radar, satellites, and high-power microwave systems generate high heat densities, and placing diamond beneath the device, or building it into a flip-chip package, helps draw heat away from the active region. In some designs, it can even replace a silicon carbide substrate outright.

High-power laser and photonics packaging are other promising applications. Diamond can be used with laser diode arrays, optical transceivers, and LiDAR systems. Thermal management matters here not just to avoid excessive heat but because even small temperature swings can shift a laser's output wavelength. So removing heat more efficiently helps keep operation stable.

How diamond is bonded into the package depends on the type of diamond being used. Single-crystal and polycrystalline diamond may require insulating or bonding layers, while a copper-diamond composite can be integrated into the existing package architecture without requiring an entirely new bonding scheme. This may sometimes need only a different surface finish or metallisation.

Diamond surfaces can be metallised with titanium, platinum, or gold to make them compatible with assembly processes, and existing materials such as silver-indium solder or conductive epoxy may still be used, particularly in thermally less demanding applications. In a